20/10/26 **Operators:** It is a special symbol used to perform certain operations between two operands.

Types of operators:

1) Arithematic Operators: Arithmetic operators are used to perform mathematical calculations.

operators are :- +, −, *, /, %, //, **

Ex: a = 10
b = 3
print (a+b):
print (a−b)
:

2) Assignment operator: Assignment operators are used to assign values to variables or update their values

operators are : =, +=, −=, *=, /=, %=, //=, **=

Ex: a = 10
a += 5
print (a)

3) Comparison operators: Comparison operators are used to compare two value. They return either True or False.

operators are : ==, !=, >, <, >=, <=

Ex: a = 10
b = 5
print (a > b)

4) Logical operators: Logical operators are used to combine conditions or reverse a condition

operators are: and, or, not.

and = True if both conditions are True

or = True if at least one condition is True

not = Reverse the result.

Ex: print (a > 5 and a < 20)
print (a < 5 or a == 10)
print (not (a > 5))

5. Bitwise operators : Bitwise operators perform operations on the binary representation of integers.

Operator : &, |, ^, ~, <<, >>

Ex : a = 5
b = 3
print(a & b)

6. Membership Operators : Membership operators check whether a value exists in a sequence such as a string, list, or tuple.

in → Return True if the value exists.

not in → Return True if the value does not exist.

Ex : fruits = ["apple", "banana", "mango"]
print ("apple" in fruits)
print ("grapes not in fruits)

O/P True
True

7. Identity Operators : Identity Operators : Identity operator check whether two variables refer to the same object in memory.

is → Returns True if both refer to the same object.
is not → Returns True if they refer to different objects.

Ex : a = [1, 2, 3]
b = a
c = [1, 2, 3]
print (a is b)
print (a is c)
print (a == c)

O/P True
False
True.